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Reference Material on Mathematics-I (MTH1001)

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Mathematics-I (MTH1001)

Unit-V

Syllabus Content:

Fourier series, Fourier expansion of functions of any period, even and odd functions.

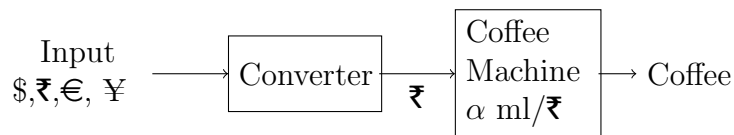
Half range expansion, Fourier transform and Fourier integral.



Fourier Series

Why Fourier ?

Let us consider the example of a coffee machine. Let the coffee machine only accepts the Indian currency. If any foreigner comes to India without rupee, then he/she may face problem, to solve the currency problem, we need to add a converter, which is explained in the following figure.



The converter will convert the different currencies into rupee. We find same type of problems in science, engineering and other areas. One such type of problem is signal processing.

In signal processing, we are searching for a converter which transforms input signals to some **currency signals**. The currency signals have the property that they can be manipulated quite easily by any given system, in particular linear systems.

The above problems can be solved with the help of Fourier Series. Our next focus is Fourier Series.

Fourier Series

Fourier Series is an infinite series representation of periodic function in terms of the trigonometric sine and cosine functions.

Most of the single valued functions which occur in applied mathematics can be expressed in the form of Fourier series, which is in terms of sines and cosines.

Fourier series is to be expressed in terms of periodic functions- sines and cosines.

Fourier series is a very powerful method to solve ordinary and partial differential equations, particularly with periodic functions appearing as non-homogeneous terms.

We know that, Taylor's series expansion is valid only for functions which are continuous and differentiable. Fourier series is possible not only for continuous functions but also for periodic functions, functions which are discontinuous in their values and derivatives. Further, because of the periodic nature, Fourier series constructed for one period is valid for all values.

Periodic Functions

A function $f(x)$ is said to be periodic function with period $T > 0$ if for all x , $f(x + T) = f(x)$, and T is the least of such values.

Ex: 1) $\sin x$, $\cos x$ are periodic functions with period 2π .

2) $\tan x$, $\cot x$ are periodic functions with period π .

Euler's Formulae

The Fourier Series for the function $f(x)$ in the interval $C \leq x \leq C + 2\pi$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where $a_0 = \frac{1}{\pi} \int_C^{C+2\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_C^{C+2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_C^{C+2\pi} f(x) \sin nx dx$$

These values a_0 , a_n , b_n are known as Euler's Formulae.



CONDITIONS FOR FOURIER EXPANSION

A function $f(x)$ defined in $[0, 2\pi]$ has a valid Fourier series expansion of the form

$$\frac{a_0}{2} + \sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Where a_0, a_n, b_n are constants, provided

- 1) $f(x)$ is well defined and single-valued, except possibly at a finite number of point in the interval $[0, 2\pi]$.
- 2) $f(x)$ has finite number of finite discontinuities in the interval in $[0, 2\pi]$.
- 3) $f(x)$ has finite number of finite maxima and minima.

Note: The above conditions are valid for the function defined in the Intervals $[-\pi, \pi], [0, 2l], [-l, l]$.

- $\{1, \cos 1x, \cos 2x, \cos 3x, \dots, \cos nx, \dots, \sin 1x, \sin 2x, \sin 3x, \dots, \sin nx, \dots\}$

Consider any two, All these have a common period 2π . Here $1 = \cos 0x$

- $\left\{1, \cos \frac{\pi x}{l}, \cos \frac{2\pi x}{l}, \cos \frac{3\pi x}{l}, \dots, \cos \frac{n\pi x}{l}, \dots, \sin \frac{\pi x}{l}, \sin \frac{2\pi x}{l}, \sin \frac{3\pi x}{l}, \dots, \sin \frac{n\pi x}{l}, \dots\right\}$

All these have a common period $2l$.

These are called complete set of orthogonal functions.

Definition of Fourier series

- Let $f(x)$ be a function defined in $[0, 2\pi]$. Let $f(x + 2\pi) = f(x) \forall x$, then the Fourier Series of

$$f(x) \text{ is given by } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

These values a_0, a_n, b_n are called as Fourier coefficients of $f(x)$ in $[0, 2\pi]$.

- Let $f(x)$ be a function defined in $[-\pi, \pi]$. Let $f(x + 2\pi) = f(x) \forall x$, then the Fourier Series of

$$f(x) \text{ is given by } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

These values a_0, a_n, b_n are called as Fourier coefficients of $f(x)$ in $[-\pi, \pi]$.



► Let $f(x)$ be a function defined in $[0, 2l]$. Let $f(x + 2l) = f(x) \forall x$, then the Fourier Series of $f(x)$ is given by $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$

$$\text{where } a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

These values a_0, a_n, b_n are called as Fourier coefficients of $f(x)$ in $[0, 2l]$.

► Let $f(x)$ be a function defined in $[-l, l]$. Let $f(x + 2\pi) = f(x) \forall x$, then the Fourier Series of $f(x)$ is given by $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$

$$\text{where } a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

These values a_0, a_n, b_n are called as Fourier coefficients of $f(x)$ in $[-l, l]$.

FOURIER SERIES FOR EVEN AND ODD FUNCTIONS

We know that if $f(x)$ be a function defined in $[-\pi, \pi]$. Let $f(x + 2\pi) = f(x) \forall x$, then the Fourier Series of $f(x)$ is given by $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$

$$\text{where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

These values a_0, a_n, b_n are called as Fourier coefficients of $f(x)$ in $[-\pi, \pi]$.

Case (i): When $f(x)$ is an even function

$$\text{then, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

Since $\cos nx$ is an even function, $f(x)$ is an even function $\Rightarrow$ Product of two even functions is even

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \quad (\because \text{Integrand is even})$$

Now, $\sin nx$ is an odd function, $f(x)$ is an even function $\Rightarrow$ Product of odd and even is odd



$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = 0 \quad (\because \text{Integrand is odd})$$

Thus, if a function $f(x)$ is even in $[-\pi, \pi]$, its Fourier series expansion contains only cosine terms.

$$\text{Hence Fourier Series is given by } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\text{where } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, n = 0, 1, 2, 3, \dots$$

Case (ii): When $f(x)$ is an Odd Function

$$\text{then, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = 0$$

Since $\cos nx$ is an even function, $f(x)$ is an odd function $\Rightarrow$ Product of even and odd is even

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = 0 \quad (\because \text{Integrand is odd})$$

Now, $\sin nx$ is an odd function, $f(x)$ is an odd function $\Rightarrow$ Product of two odd functions is even

$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx \quad (\because \text{Integrand is even})$$

Thus, if a function $f(x)$ is Odd in $[-\pi, \pi]$, its Fourier series expansion contains only sine terms.

Hence, if $f(x)$ is odd function defined in $[-\pi, \pi]$, $f(x)$ can be expanded as a series of the form

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{where, } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

HALF RANGE FOURIER SERIES

Half Range Fourier Sine Series defined in $[0, \pi]$: The Fourier half range sine series in $[0, \pi]$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx,$$

$$\text{where, } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

This is Similar to the Fourier series defined for odd function in $[-\pi, \pi]$



Half Range Fourier Cosine Series defined in $[0, \pi]$: The Fourier half range Cosine series in $[0, \pi]$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where, $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

This is Similar to the Fourier series defined for even function in $[-\pi, \pi]$

Half Range Fourier Sine Series defined in $[0, l]$: The Fourier half range sine series in $[0, \pi]$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

where, $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

This is Similar to the Fourier series defined for odd function in $[-l, l]$

Half Range Fourier Cosine Series defined in $[0, l]$: The Fourier half range Cosine series in $[0, l]$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

where, $a_0 = \frac{2}{l} \int_0^l f(x) dx$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

This is Similar to the Fourier series defined for even function in $[-l, l]$

Important Formulae

$$\blacktriangleright \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \\ 0 & \text{if } f(2a-x) = -f(x) \end{cases}$$

$$\blacktriangleright \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f \text{ is even} \\ 0 & \text{if } f \text{ is odd} \end{cases}$$

Here Even function means: If $f(-x) = f(x)$, then $f(x)$ is called as even function

Odd function means : If $f(-x) = -f(x)$, then $f(x)$ is called as odd function.

$$\blacktriangleright \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\blacktriangleright \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx ; a < c < b$$

$$\blacktriangleright \int_0^a f(x) dx = \int_0^a f(a-x) dx, \text{ Also } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

Problems on Fourier Series

1) Find the Fourier series to represent $f(x) = x^2$ in the interval $(0, 2\pi)$.

Sol: We know that, the Fourier series of $f(x)$ defined in the interval $(0, 2\pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where, $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Here, $f(x) = x^2$

Now, $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} x^2 dx$

$$= \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{2\pi} = \frac{1}{3\pi} [(2\pi)^3 - 0] = \frac{8}{3} \pi^2$$

$$\Rightarrow a_0 = \frac{8}{3} \pi^2$$

Again, $a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{2\pi} \underbrace{x^2}_u \underbrace{\cos nx}_v dx$

$$= \frac{1}{\pi} \left[x^2 \int \cos nx dx - \left\{ \int \frac{d}{dx} (x^2) (\int \cos nx dx) dx \right\} \right]$$

$$\left[\because \int uv dx = u \int v dx - \left\{ \int \frac{du}{dx} \cdot (\int v dx) dx \right\} \right]$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - \left\{ \int 2x \left(\frac{\sin nx}{n} \right) dx \right\} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - \frac{2}{n} \left\{ \int \underbrace{x}_u \underbrace{\sin nx}_v dx \right\} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - \frac{2}{n} \left(-x \frac{\cos nx}{n} + \int 1 \cdot \frac{\cos nx}{n} dx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - \frac{2}{n} \left(-x \frac{\cos nx}{n} + \frac{1}{n} \int \cos nx dx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - \frac{2}{n} \left(-x \frac{\cos nx}{n} + \frac{1}{n} \frac{\sin nx}{n} \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) + \frac{2}{n^2} x \cos nx - \frac{2}{n^3} \sin nx \right]_0^{2\pi}$$

$$= \frac{4}{n^2} \left[\because \cos 2n\pi = 1 \right]$$



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$$\Rightarrow a_n = \frac{4}{n^2}$$

$$\text{Again, } b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \int_0^{2\pi} \underbrace{x^2}_u \underbrace{\sin nx}_v \, dx$$

$$= \frac{1}{\pi} \left[x^2 \int \sin nx \, dx - \left\{ \int \frac{d}{dx} (x^2) (\int \sin nx \, dx) dx \right\} \right]$$

$$\left[\because \int uv \, dx = u \int v \, dx - \left\{ \int \frac{du}{dx} \cdot (\int v \, dx) dx \right\} \right]$$

$$= \frac{1}{\pi} \left[x^2 \left(-\frac{\cos nx}{n} \right) - \left\{ \int 2x \left(-\frac{\cos nx}{n} \right) dx \right\} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-x^2 \left(\frac{\cos nx}{n} \right) + \frac{2}{n} \left\{ \int \underbrace{x}_u \underbrace{\cos nx}_v \, dx \right\} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-x^2 \left(\frac{\cos nx}{n} \right) + \frac{2}{n} \left(x \frac{\sin nx}{n} + \int 1 \cdot \frac{\sin nx}{n} \, dx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-x^2 \left(\frac{\cos nx}{n} \right) + \frac{2}{n} \left(x \frac{\sin nx}{n} + \frac{1}{n} \int \sin nx \, dx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-x^2 \left(\frac{\cos nx}{n} \right) + \frac{2}{n} \left(x \frac{\sin nx}{n} + \frac{1}{n} \frac{\cos nx}{n} \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-x^2 \left(\frac{\cos nx}{n} \right) + \frac{2}{n^2} x \sin nx + \frac{2}{n^3} \cos nx \right]_0^{2\pi}$$

$$= -\frac{4\pi}{n} \left[\because \begin{matrix} \cos 2n\pi = 1 \\ \sin 2n\pi = 0 \end{matrix} \right]$$

$$\Rightarrow b_n = -\frac{4\pi}{n}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\therefore f(x) = x^2 = \frac{8\pi^2}{2} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

$$\Rightarrow x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

This is the Fourier series for the function $f(x) = x^2$

Hence the result



2) Find the Fourier series of the periodic function defined as $f(x) = \begin{cases} -\pi & ; -\pi < x < 0 \\ x & ; 0 < x < \pi \end{cases}$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Sol: We know that, the Fourier series of $f(x)$ defined in the interval $(-\pi, \pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$\text{Here, } f(x) = \begin{cases} -\pi & ; -\pi < x < 0 \\ x & ; 0 < x < \pi \end{cases}$$

$$\begin{aligned} \text{Now, } a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) dx + \int_0^{\pi} f(x) dx \right] \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) dx + \int_0^{\pi} x dx \right] \\ &= \frac{1}{\pi} \left[(-\pi) \int_{-\pi}^0 dx + \int_0^{\pi} x dx \right] \\ &= \frac{1}{\pi} \left[(-\pi) [x]_{-\pi}^0 + \left[\frac{x^2}{2} \right]_0^{\pi} \right] = \frac{1}{\pi} \left[(-\pi)(\pi) + \frac{\pi^2}{2} \right] \\ &= \frac{1}{\pi} \left[-\pi^2 + \frac{\pi^2}{2} \right] = -\frac{\pi}{2} \\ &\Rightarrow \boxed{a_0 = -\frac{\pi}{2}} \end{aligned}$$

$$\begin{aligned} \text{Also, } a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \cos nx dx + \int_0^{\pi} f(x) \cos nx dx \right] \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) \cos nx dx + \int_0^{\pi} x \cos nx dx \right] \\ &= \frac{1}{\pi} \left[-\pi \int_{-\pi}^0 (\cos nx) dx + \int_0^{\pi} x \cos nx dx \right] \\ &= \frac{1}{\pi} \left[-\pi \left(\frac{\sin nx}{n} \right)_{-\pi}^0 + \left\{ x \left(\frac{\sin nx}{n} \right) - \int 1 \left(\frac{\sin nx}{n} \right) dx \right\}_0^{\pi} \right] \\ &= \frac{1}{\pi} \left[-\frac{\pi}{n} (\sin nx)_{-\pi}^0 + \left\{ \frac{x \sin nx}{n} - \frac{1}{n} \int \sin nx dx \right\}_0^{\pi} \right] \\ &= \frac{1}{\pi} \left[-\frac{\pi}{n} (\sin nx)_{-\pi}^0 + \left\{ \frac{x \sin nx}{n} - \frac{1}{n} \left(\frac{-\cos nx}{n} \right) \right\}_0^{\pi} \right] \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{\pi} \left[-\frac{\pi}{n} (\sin nx)^0_{-\pi} + \left\{ \frac{x \sin nx}{n} + \frac{1}{n^2} \cos nx \right\}_0^{\pi} \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi}{n} [0 - \sin(-n\pi)] + \left\{ \left(\frac{\pi \sin n\pi}{n} + \frac{1}{n^2} \cos n\pi \right) - \left(0 + \frac{1}{n^2} \cos n0 \right) \right\} \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi}{n} \sin n\pi + \left\{ \left(\frac{\pi \sin n\pi}{n} + \frac{1}{n^2} \cos n\pi \right) - \frac{1}{n^2} \cdot 1 \right\} \right] \quad \left[\begin{array}{l} \because \cos(-\theta) = \cos \theta \\ \sin(-\theta) = -\sin \theta \end{array} \right] \\
 \Rightarrow a_n &= \frac{1}{\pi} \left[-\frac{\pi}{n} (0) + \left\{ \left(\frac{\pi(0)}{n} + \frac{1}{n^2} (-1)^n \right) - \frac{1}{n^2} \right\} \right] \quad \left[\begin{array}{l} \because \sin n\pi = 0 \\ \text{&} \\ \cos n\pi = (-1)^n \end{array} \right] \\
 &= \frac{1}{\pi} \left[\frac{(-1)^n}{n^2} + \frac{1}{n^2} \right] = \frac{1}{\pi n^2} [(-1)^n - 1] \\
 \Rightarrow a_n &= \frac{1}{\pi n^2} [(-1)^n - 1]
 \end{aligned}$$

Again, $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$

$$\begin{aligned}
 &= \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \sin nx \, dx + \int_0^{\pi} f(x) \sin nx \, dx \right] \\
 &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) \sin nx \, dx + \int_0^{\pi} x \sin nx \, dx \right] \\
 &= \frac{1}{\pi} \left[-\pi \int_{-\pi}^0 (\sin nx) \, dx + \int_0^{\pi} x \sin nx \, dx \right] \\
 &= \frac{1}{\pi} \left[-\pi \left(-\frac{\cos nx}{n} \right)_{-\pi}^0 + \left\{ x \left(-\frac{\cos nx}{n} \right) - \int 1 \left(-\frac{\cos nx}{n} \right) dx \right\}_0^{\pi} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} (\cos nx)^0_{-\pi} + \left\{ -\frac{x \cos nx}{n} + \frac{1}{n} \int \cos nx \, dx \right\}_0^{\pi} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} (\cos nx)^0_{-\pi} + \left\{ -\frac{x \cos nx}{n} + \frac{1}{n} \frac{(\sin nx)}{n} \right\}_0^{\pi} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} (\cos nx)^0_{-\pi} + \left\{ -\frac{x \cos nx}{n} + \frac{1}{n^2} \sin nx \right\}_0^{\pi} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} [1 - \cos(-n\pi)] + \left\{ \left(-\frac{\pi \cos n\pi}{n} + \frac{1}{n^2} \sin n\pi \right) - \left(-0 + \frac{1}{n^2} \sin n0 \right) \right\} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} [1 - \cos n\pi] - \frac{\pi \cos n\pi}{n} \right] = \frac{1}{n} (1 - 2 \cos n\pi) \\
 &\quad \left[\begin{array}{l} \because \cos(-\theta) = \cos \theta \quad \sin n\pi = 0 \\ \sin(-\theta) = -\sin \theta \quad \text{&} \quad \cos n\pi = (-1)^n \end{array} \right] \\
 \Rightarrow b_n &= \frac{1}{n} (1 - 2 \cos n\pi)
 \end{aligned}$$

Hence, the Fourier series for given $f(x)$ is given by

$$f(x) = \frac{-\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{1}{n} (1 - 2 \cos n\pi) \sin nx \right)$$



$$\Rightarrow f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty} \left(\frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{1}{n} (1 - 2 \cos n\pi) \sin nx \right)$$

$$\Rightarrow f(x) = -\frac{\pi}{4} - \frac{2}{\pi} \left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right) + \left(3 \sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \dots \right)$$

Deduction: Put $x = 0$ in the above function $f(x)$, we get

$$f(0) = -\frac{\pi}{4} - \frac{2}{\pi} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

Since, $f(x)$ is discontinuous at $x = 0$, $f(0-0) = -\pi$
 $f(0+0) = 0$

$$\Rightarrow f(0) = \frac{1}{2} [f(0-0) + f(0+0)]$$

$$\Rightarrow f(0) = \frac{1}{2} (-\pi) = -\frac{\pi}{2}$$

$$\text{Hence, } f(0) = -\frac{\pi}{2} = -\frac{\pi}{4} - \frac{2}{\pi} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

$$\Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

Hence the result

3) Expand the function $f(x) = x^2$ as Fourier series in $[-\pi, \pi]$.

Hence deduce that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$

Sol: We know that, the Fourier series of $f(x)$ defined in the interval $(-\pi, \pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Here, $f(x) = x^2$

$$\text{Now, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$



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$$\Rightarrow a_0 = \frac{2\pi^2}{3}$$

$$\text{Again, } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx \quad [\because f(x) \text{ is even} \Rightarrow \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx]$$

$$= \frac{2}{\pi} \left[\frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2x \sin nx}{n^3} \right]_0^{\pi} = \frac{4}{n^2} (-1)^n$$

$$\Rightarrow a_n = \frac{4}{n^2} (-1)^n$$

$$\text{Again, } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx \, dx$$

$$= 0 \quad [\because f(x) \text{ is odd} \Rightarrow \int_{-a}^a f(x) dx = 0]$$

Hence, the Fourier series for given $f(x)$ is given by

$$f(x) = x^2 = \frac{\left(\frac{2\pi^2}{3}\right)}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$$

$$\Rightarrow x^2 = \frac{\pi^2}{3} + 4 \left(-\cos x + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \frac{\cos 4x}{4^2} - \dots \right)$$

Deduction: Put $x = \pi$ in the above equation, we get

$$\Rightarrow \pi^2 = \frac{\pi^2}{3} + 4 \left(-\cos \pi + \frac{\cos 2\pi}{2^2} - \frac{\cos 3\pi}{3^2} + \frac{\cos 4\pi}{4^2} - \dots \right)$$

$$\Rightarrow \pi^2 - \frac{\pi^2}{3} = 4 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \right)$$

$$\Rightarrow \frac{2\pi^2}{3} = 4 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \right)$$

$$\Rightarrow \frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

Hence the Result



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4) Expand the function

$$f(x) = \begin{cases} 0 & (-\pi < x < 0) \\ \pi - x & (0 \leq x < +\pi) \end{cases} \quad \text{in a Fourier series.}$$

$$L = \pi.$$

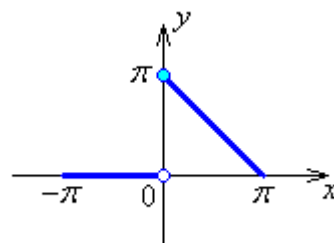
$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} (\pi - x) dx \\ &= 0 + \frac{1}{\pi} \left[\frac{(\pi - x)^2}{-2} \right]_0^{\pi} = \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = 0 + \frac{1}{\pi} \int_0^{\pi} (\pi - x) \cos nx dx \\ &= \frac{1}{\pi} \left[\frac{n(\pi - x) \sin nx - \cos nx}{n^2} \right]_0^{\pi} = \frac{1 - (-1)^n}{n^2 \pi} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0 + \frac{1}{\pi} \int_0^{\pi} (\pi - x) \sin nx dx \\ &= \frac{1}{\pi} \left[\frac{n(\pi - x) \cos nx + \sin nx}{-n^2} \right]_0^{\pi} = \frac{1}{n} \end{aligned}$$

Therefore the Fourier series for $f(x)$ is

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left(\frac{1 - (-1)^n}{n^2 \pi} \cos nx + \frac{1}{n} \sin nx \right) \quad (-\pi < x < +\pi)$$



D	I
$\pi - x$	$\cos nx$
+	
-1	$\frac{1}{n} \sin nx$
-	
0	$-\frac{1}{n^2} \cos nx$

D	I
$\pi - x$	$\sin nx$
+	
-1	$-\frac{1}{n} \cos nx$
-	
0	$-\frac{1}{n^2} \sin nx$



Fourier Integral and Transform

- The central starting point of Fourier analysis is **Fourier series**. They are infinite series designed to represent general periodic functions in terms of simple ones, namely, cosines and sines.
- This trigonometric system is ***orthogonal***, allowing the computation of the coefficients of the Fourier series by use of the well-known Euler formulas, as shown. Fourier series are very important to the engineer and physicist because they allow the solution of linear differential equations and partial differential.
- Fourier series are, in a certain sense, more universal than the familiar Taylor series in calculus because many *discontinuous* periodic functions that come up in applications can be developed in Fourier series but do not have Taylor series expansions.



- The Fourier Transform is a tool that breaks a waveform (a function or signal) into an alternate representation, characterized by sine and cosines. The Fourier Transform shows that any waveform can be re-written as the sum of sinusoidal functions.
- The Fourier transform is a mathematical function that decomposes a waveform, which is a function of time, into the frequencies that make it up. The result produced by the Fourier transform is a complex valued function of frequency.
- The absolute value of the Fourier transform represents the frequency value present in the original function and its complex argument represents the phase offset of the basic sinusoidal in that frequency.



- The Fourier transform is also called a generalization of the Fourier series. This term can also be applied to both the frequency domain representation and the mathematical function used.
- The Fourier transform helps in extending the Fourier series to non-periodic functions, which allows viewing any function as a sum of simple sinusoids.



OBJECTIVES

- After studying this chapter we will learn about how Fourier transforms is useful many physical applications, such as partial differential equations and heat transfer equations.
- With the use of different properties of Fourier transform along with Fourier sine transform and Fourier cosine transform, one can solve many important problems of physics with very simple way.
- Thus we will learn from this unit to use the Fourier transform for solving many physical application related partial differential equations.



SOME IMPORTANT RESULTS

- $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$
- $\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$
- $\int_0^\infty \frac{\sin ax}{x} \, dx = \frac{\pi}{2}$
- $\int_0^\infty e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$
- $\int_{-\infty}^\infty \frac{\sin mx}{(x-b)^2 + a^2} \, dx = \frac{\pi}{a} e^{-am} \sin bm, \quad [m > 0]$



FOURIER INTEGRAL

- Fourier series are powerful tools for problems involving functions that are periodic or are of interest on a finite interval only.
- Since, of course, many problems involve functions that are ***nonperiodic and are of interest on the whole x -axis***, we ask what can be done to extend the method of Fourier series to such functions. This idea will lead to “Fourier integrals.”



FOURIER INTEGRAL THEOREM

Fourier integral theorem states that $f(x) = \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(t) \cos u(t-x) dt du$

Proof. We know that Fourier series of a function $f(x)$ in $(-c, c)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^\infty a_n \cos \frac{n\pi x}{c} + \sum_{n=1}^\infty b_n \sin \frac{n\pi x}{c}$$

Where a_0 , a_n and b_n are given by

$$\begin{aligned} a_0 &= \frac{1}{c} \int_{-c}^c f(t) dt, \\ a_n &= \frac{1}{c} \int_{-c}^c f(t) \cos \frac{n\pi t}{c} dt, \\ b_n &= \frac{1}{c} \int_{-c}^c f(t) \sin \frac{n\pi t}{c} dt, \end{aligned}$$

Substituting the values of a_0 , a_n and b_n in above equation, we get

$$f(x) = \frac{1}{2c} \int_{-c}^c f(t) dt + \sum_{n=1}^\infty \frac{1}{c} \int_{-c}^c f(t) \cos \frac{n\pi t}{c} dt \cos \frac{n\pi x}{c} + \sum_{n=1}^\infty \frac{1}{c} \int_{-c}^c f(t) \sin \frac{n\pi t}{c} dt \sin \frac{n\pi x}{c}$$



$$f(x) = \frac{1}{2c} \int_{-c}^c f(t) dt + \sum_{n=1}^{\infty} \frac{1}{c} \int_{-c}^c f(t) \left[\cos \frac{n\pi t}{c} \cos \frac{n\pi x}{c} + \sin \frac{n\pi t}{c} \sin \frac{n\pi x}{c} \right] dt$$

$$f(x) = \frac{1}{2c} \int_{-c}^c f(t) dt + \sum_{n=1}^{\infty} \frac{1}{c} \int_{-c}^c f(t) \left[\cos \left(\frac{n\pi t}{c} - \frac{n\pi x}{c} \right) \right] dt$$

$$f(x) = \frac{1}{2c} \int_{-c}^c f(t) dt + \sum_{n=1}^{\infty} \frac{1}{c} \int_{-c}^c f(t) \left[\cos \frac{n\pi}{c} (t - x) \right] dt$$

$$f(x) = \frac{1}{2c} \int_{-c}^c f(t) \left\{ 1 + 2 \sum_{n=1}^{\infty} \cos \frac{n\pi}{c} (t - x) \right\} dt$$

Since cosine functions are even functions i.e., $\cos(-\theta) = \cos \theta$ the expression

$$\left\{ 1 + 2 \sum_{n=1}^{\infty} \cos \frac{n\pi}{c} (t - x) \right\} = \sum_{n=-\infty}^{\infty} \cos \frac{n\pi}{c} (t - x)$$



We now let the parameter c approach infinity, transforming the finite interval $[-c, c]$ into the infinite interval $(-\infty$ to $+\infty)$. We set

$$\frac{n\pi}{c} = \omega, \text{ and } \frac{\pi}{c} = d\omega \quad \text{with } c \rightarrow \infty$$

Then we have

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left\{ \int_{-\infty}^{\infty} d\omega \cos \omega(t - x) \right\} dt$$

On simplifying

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \omega(t - x) d\omega dt \quad \text{Proved}$$



FOURIER SINE AND COSINE INTEGRALS

$$f(x) = \frac{2}{\pi} \int_0^\infty \sin \omega x \, d\omega \int_0^\infty f(t) \sin \omega t \, dt \quad (\text{Fourier Sine Integrals})$$

$$f(x) = \frac{2}{\pi} \int_0^\infty \cos \omega x \, d\omega \int_0^\infty f(t) \cos \omega t \, dt \quad (\text{Fourier Cosine Integrals})$$

Proof: We can write

$$\cos \omega(t - x) = \cos(\omega t - \omega x) = \cos \omega t \cos \omega x + \sin \omega t \sin \omega x$$

Using this expansion in Fourier integral theorem, we have

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty \cos \omega(t - x) \, d\omega \, dt \\ \Rightarrow f(x) &= \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(t) (\cos \omega t \cos \omega x + \sin \omega t \sin \omega x) \, d\omega \, dt \\ \Rightarrow f(x) &= \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(t) \cos \omega t \cos \omega x \, d\omega \, dt + \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(t) \sin \omega t \sin \omega x \, d\omega \, dt \end{aligned}$$



Now to solve the above equation, we have two different cases, using the following conditions

$$\int_{-a}^a f(x) dx = 0 \quad \text{for odd function}$$

And

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \quad \text{for even function}$$

Case I: when $f(t)$ is even function: this means

$$\Rightarrow f(t) \sin \omega t \quad \text{is odd function and} \\ f(t) \cos \omega t \quad \text{is even function}$$

Hence

$$\frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \sin \omega t \sin \omega x d\omega dt = 0$$

And

$$\Rightarrow f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) (\cos \omega t \cos \omega x d\omega dt = \frac{2}{\pi} \int_0^{\infty} \cos \omega x d\omega \int_{-\infty}^{\infty} f(t) \cos \omega t dt$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos \omega x du \int_0^{\infty} f(t) \cos \omega t dt$$

This is known as Fourier cosine integral.



Case II: If $f(t)$ is odd function: this means

$\Rightarrow f(t) \sin \omega t$ is even function and
 $f(t) \cos \omega t$ is odd function

Hence

$$\frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \omega t \cos \omega x d\omega dt = 0$$

And

$$\begin{aligned} \Rightarrow f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \sin \omega t \sin \omega x d\omega dt = \frac{2}{\pi} \int_0^{\infty} \sin \omega x d\omega \int_{-\infty}^{\infty} f(t) \sin \omega t dt \\ f(x) &= \frac{2}{\pi} \int_0^{\infty} \sin \omega x du \int_0^{\infty} f(t) \sin \omega t dt \end{aligned}$$

This is known as Fourier sine integral.



FOURIER'S COMPLEX INTEGRALS

We know from Fourier integral theorem

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) d\omega dt$$

Now adding

$$f(x) = \frac{i}{2\pi} \int_{-\infty}^{\infty} f(t) dt \int_{-\infty}^{\infty} \sin \omega(t-x) d\omega = 0$$

Since

$$\int_{-\infty}^{\infty} \sin \omega(t-x) d\omega = 0 \quad \text{because of odd function}$$

Hence

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) d\omega dt + \frac{i}{2\pi} \int_{-\infty}^{\infty} f(t) dt \int_{-\infty}^{\infty} \sin \omega(t-x) d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) dt \left[\int_{-\infty}^{\infty} \cos \omega(t-x) + i \sin \omega(t-x) \right] d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) dt \left[\int_{-\infty}^{\infty} e^{i\omega(t-x)} \right] d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} d\omega \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$$

This relation is known as Fourier's complex Integral.



Example 5. Express the following function

$$f(x) = \begin{cases} 1 & \text{when } x \leq 1 \\ 0 & \text{when } x > 1 \end{cases}$$

as a Fourier integral. Hence evaluate

$$\int_0^{\infty} \frac{\sin u \cos ux}{u} du$$

Solution: we know the Fourier Integral theorem, the Fourier Integral of a function $f(x)$ is given by

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) d\omega dt$$

Using $\omega = u$ we have

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos u(t-x) du dt \\ f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_{-1}^1 \cos u(t-x) dt du \quad \text{since } f(t) = 1 \end{aligned}$$



Now integrating w.r.t. t we have

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\frac{\sin u(t-x)}{u} \right]_{-1}^1 du$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\frac{\sin u(1-x) + \sin u(1+x)}{u} \right] du$$

Now using $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$ and solving it we will get

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin u \cos ux}{u} du$$

We can rewrite this

$$\int_0^{\infty} \frac{\sin u \cos ux}{u} du = \frac{\pi}{2} f(x)$$

$$\int_0^{\infty} \frac{\sin u \cos ux}{u} du = \begin{cases} \frac{\pi}{2} \times 1 = \frac{\pi}{2}, & \text{for } x < 1 \\ \frac{\pi}{2} \times 0 = 0, & \text{for } x > 1 \end{cases}$$

For $x=1$, which is a point of discontinuity of $f(x)$, value of integral $= \frac{\frac{\pi}{2} + 0}{2} = \frac{\pi}{4}$



FOURIER TRANSFORMS

From the Fourier complex integral we know that

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} d\omega \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$$

We can rewrite the above expression as follows using $\omega = s$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-isx} ds \int_{-\infty}^{\infty} f(t) e^{ist} dt = \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isx} ds \right] \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} dt \right]$$



Now using $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} dt = F(s)$ in above equation, we get

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isx} F(s) ds$$

Where $F(s)$ is called the Fourier Transform of $f(x)$.

And $f(x)$ is called the Inverse Fourier transform of $F(s)$.

Thus , we obtain the definition of Fourier transform is

$$F(s) = F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} dt$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isx} \cdot F(s) ds$$



FOURIER SINE TRANSFORMS

We know that from Fourier sine integral

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin sx \, ds \int_0^{\infty} f(t) \sin st \, dt = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sin sx \, ds \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin st \, dt \right]$$

$$\text{Now putting } F(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin st \, dt$$

We have

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sin sx \, ds F(s)$$

In above equation $F(s)$ is called Fourier Sine transform of $f(x)$

$$F(s) = F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin st \, dt$$

And $f(x)$ given below is known as inverse Fourier Sine transform of $F(s)$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F(s) \sin sx \, ds$$



FOURIER COSINE TRANSFORM

From Fourier cosine integral we know that

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos \omega x \, d\omega \int_0^{\infty} f(t) \cos \omega t \, dt$$
$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \cos sx \, ds \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos st \, dt \right]$$

Now putting $F(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos st \, dt$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \cos sx \, ds F(s)$$

In above equation $F(s)$ is called Fourier cosine transform of $f(x)$

$$\mathbf{F(s) = F_c [f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos st \, dt}$$

And $f(x)$ given below is known as inverse Fourier cosine transform of $F(s)$

$$\mathbf{f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \cos sx F(s) ds}$$



Example 6: Find the Fourier transform of e^{-ax^2} , where $a > 0$.

Solution : The Fourier transform of $f(x)$:

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Hence

$$\begin{aligned} F\{e^{-ax^2}\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} e^{isx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2 + isx} dx \\ \Rightarrow F\{e^{-ax^2}\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2 - \frac{s^2}{4a} + isx + \frac{s^2}{4a}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(x\sqrt{a} - \frac{is}{2\sqrt{a}}\right)^2 - \frac{s^2}{4a}} dx \\ \Rightarrow F\{e^{-ax^2}\} &= \frac{e^{-\frac{s^2}{4a}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(x\sqrt{a} - \frac{is}{2\sqrt{a}}\right)^2} dx \end{aligned}$$

Putting $x\sqrt{a} - \frac{is}{2\sqrt{a}} = u \Rightarrow dx = \frac{du}{\sqrt{a}}$ in above expression we get,

$$\begin{aligned} \Rightarrow F\{e^{-ax^2}\} &= \frac{e^{-\frac{s^2}{4a}}}{\sqrt{2\pi a}} \int_{-\infty}^{\infty} e^{-u^2} du \quad \left[\text{since } \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \right] \\ \Rightarrow F\{e^{-ax^2}\} &= \frac{e^{-\frac{s^2}{4a}}}{\sqrt{2\pi a}} \sqrt{\pi} = \frac{e^{-\frac{s^2}{4a}}}{\sqrt{2a}} \quad \text{Ans.} \end{aligned}$$



Example 7: Find the Fourier transform of

$$f(x) = \begin{cases} 2 & \text{for } |x| < a \\ 0 & \text{for } |x| > a \end{cases}$$

Solution: We know that the Fourier transform of a function is given by

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Using the given value of $f(x)$ we get,

$$\begin{aligned} F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a 2e^{isx} dx = \frac{2}{\sqrt{2\pi}} \int_{-a}^a e^{isx} dx = \\ F\{f(x)\} &= \frac{2}{\sqrt{2\pi}} \left[\frac{e^{isx}}{is} \right]_{-a}^a = \frac{2}{\sqrt{2\pi} is} [e^{ias} - e^{-ias}] = \frac{4}{\sqrt{2\pi} s} \frac{[e^{ias} - e^{-ias}]}{2i} \\ F\{f(x)\} &= \frac{4}{\sqrt{2\pi} s} \sin as = 2 \sqrt{\frac{2}{\pi}} \frac{\sin sa}{s} \quad \text{Ans.} \end{aligned}$$



Example 8: Find Fourier Sine transform of $\frac{1}{x}$.

Solution: We have to find the Fourier sine transform of $f(x) = \frac{1}{x}$

We know that from Fourier sine transform

$$F_s [f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx$$

Now using the value of $f(x) = \frac{1}{x}$, we get,

$$F_s [f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{1}{x} \sin sx \, dx$$

$$\text{now using } sx = t \Rightarrow dx = \frac{dt}{s}$$

We get

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\sin t}{t} \, dt = \sqrt{\frac{2}{\pi}} \left(\frac{\pi}{2} \right) \Rightarrow \text{since } \int_0^{\infty} \frac{\sin t}{t} \, dt = \frac{\pi}{2}$$

Hence

$$F_s [f(x)] = \sqrt{\frac{\pi}{2}} \quad \text{Ans.}$$



Example 9: Find the Fourier Sine Transform of e^{-ax} .

Solution: Here, $f(x) = e^{-ax}$.

The Fourier sine transform of $f(x)$:

$$F_S [f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx$$

On putting the value of $f(x)$ in (1), we get

$$F_S [e^{-ax}] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \sin sx \, dx$$

On Integrating by parts, we get

$$\begin{aligned} F_S [e^{-ax}] &= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-ax}}{a^2 + s^2} [-a \sin sx - s \cos sx] \right]_0^{\infty} \\ \text{using } \left[\int_0^{\infty} e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right] \\ &= \sqrt{\frac{2}{\pi}} \left[0 - \frac{1}{a^2 + s^2} (-s) \right] = \sqrt{\frac{2}{\pi}} \left(\frac{s}{a^2 + s^2} \right) \quad \text{Ans.} \end{aligned}$$



Example 10: Find the Fourier Cosine Transform of $f(x) = 5e^{-2x} + 2e^{-5x}$

Solution: The Fourier Cosine Transform of $f(x)$ is given by

$$F_c \{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$$

Putting the value of $f(x)$, we get

$$\begin{aligned} F_c \{f(x)\} &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} (5e^{-2x} + 2e^{-5x}) \cos sx \, dx \\ &= 5 \int_0^{\infty} e^{-2x} \cos sx \, dx + 2 \int_0^{\infty} e^{-5x} \cos sx \, dx \\ &\quad \text{using } \left[\int_0^{\infty} e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) \right] \\ &= 5 \left[\frac{e^{-2x}}{(-2)^2 + s^2} (-2 \cos sx + s \sin sx) \right]_0^{\infty} + 2 \left[\frac{e^{-5x}}{(-5)^2 + s^2} (-5 \cos sx + s \sin sx) \right]_0^{\infty} \\ &= 5 \left[0 - \frac{1}{4 + s^2} (-2) \right] + 2 \left[0 - \frac{1}{25 + s^2} (-5) \right] = 5 \left(\frac{2}{s^2 + 4} \right) + 2 \left(\frac{5}{s^2 + 25} \right) \\ &= 10 \left(\frac{1}{s^2 + 4} + \frac{1}{s^2 + 25} \right) \quad \text{Ans.} \end{aligned}$$



PROPERTIES OF FOURIER TRANSFORMS

LINEAR PROPERTY : If $F_1(s)$ and $F_2(s)$ are Fourier transforms of $f_1(x)$ and $f_2(x)$ respectively then

$$F[af_1(x) + b f_2(x)] = a F_1(s) + b F_2(s) \quad \text{where } a \text{ and } b \text{ are constants.}$$

Proof: we know from the definition of Fourier transform

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

We can write

$$F_1(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(x) e^{isx} dx$$

And

$$F_2(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_2(x) e^{isx} dx$$

Now

$$F[af_1(x) + b f_2(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [af_1(x) + b f_2(x)] e^{isx} dx$$

$$= a \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_1(x) e^{isx} dx + b \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f_2(x) e^{isx} dx$$

$$\Rightarrow F[af_1(x) + b f_2(x)] = a F_1(s) + b F_2(s) \quad \text{Proved}$$



CHANGE OF SCALE PROPERTY

We know that Fourier transform equation is given by

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Then

$$F\{f(ax)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

Proof: we know

$$\begin{aligned} F(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\ \Rightarrow F\{f(ax)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) e^{isx} dx \quad \left[\text{now put } ax = t \Rightarrow dx = \frac{dt}{a} \right] \end{aligned}$$

We have

$$\begin{aligned} F\{f(ax)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\frac{s}{a}t} \frac{dt}{a} = \frac{1}{a} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\left(\frac{s}{a}\right)t} dt \\ &\Rightarrow F\{f(ax)\} = \frac{1}{a} F\left(\frac{s}{a}\right) \quad \text{Proved} \end{aligned}$$



SHIFTING PROPERTY

The Fourier transform equation is given by

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Then

$$F\{f(x - a)\} = e^{isa} F(s)$$

Proof: Given

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

then

$$F\{f(x - a)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x - a) e^{isx} dx$$

$$\text{Put } (x - a) = u \Rightarrow x = u + a \text{ and } dx = du$$

We have

$$F\{f(x - a)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{is(u+a)} du = e^{isa} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{isu} du$$

$$\Rightarrow F\{f(x - a)\} = e^{isa} F(s)$$

Proved



FOURIER TRANSFORM OF DERIVATIVES

As we know from the properties of Fourier Transform

$$F\{f^n(x)\} = (-i s)^n F(s)$$

$$F\left(\frac{\partial^2 f}{dx^2}\right) = (-i s)^2 F\{f(x)\} = -s^2 \bar{f} \quad [\text{where } \bar{f} \text{ is Fourier Transform of } f]$$

If F_c and F_s are cosine and sine Fourier transform $f(x)$ then

$$F_c\{f'(x)\} = -\sqrt{\frac{2}{\pi}} f(0) + s F_s(s)$$

Proof: From cosine Fourier transform we know that

$$F_c[f'(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \cos sx \, dx = \sqrt{\frac{2}{\pi}} \int_0^\infty \cos sx \, d\{f(x)\}$$



Now integrating by parts, we get

$$\begin{aligned} &= \sqrt{\frac{2}{\pi}} [\cos sx \ f(x)]_0^{\infty} - \sqrt{\frac{2}{\pi}} \left[-s \int_0^{\infty} \sin sx \ f(x) \ dx \right] \\ &= \sqrt{\frac{2}{\pi}} [0 - f(0)] + s \sqrt{\frac{2}{\pi}} \left[\int_0^{\infty} \sin sx \ f(x) \ dx \right] \quad \{ \text{assuming } f(x) \rightarrow 0 \text{ as } x \rightarrow \infty \} \end{aligned}$$

Hence

$$F_c\{f'(x)\} = -\sqrt{\frac{2}{\pi}} f(0) + sF_s(s) \quad \text{where } \sqrt{\frac{2}{\pi}} \left[\int_0^{\infty} \sin sx \ f(x) \ dx \right] = F_s(s)$$



FOURIER TRANSFORM OF

PARTIAL DERIVATIVE OF A FUNCTION

The Fourier transform of the partial derivatives is given by

$$F \left[\frac{\partial^2 u}{\partial^2 x} \right] = -s^2 F(u)$$

Where $F(u)$ is the Fourier transform of u .

The Fourier sine transform of the partial derivatives is given by

$$F_s \left[\frac{\partial^2 u}{\partial^2 x} \right] = s(u)_{x=0} - s^2 F_s(u)$$

Where $F_s(u)$ is the Fourier sine transform of u

The Fourier cosine transform of the partial derivatives is given by

$$F_c \left[\frac{\partial^2 u}{\partial^2 x} \right] = - \left[\frac{\partial u}{\partial x} \right]_{x=0} - s^2 F_c(u)$$

Where $F_c(u)$ is the Fourier cosine transform of u .



Extra Questions

1) Find the Fourier Transform of $f(x)$ if

$$f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$$

2) Show that the Fourier Transform of

$$f(x) = \begin{cases} a - |x| & \text{for } |x| < a \\ 0 & \text{for } |x| > a > 0 \end{cases}$$

$$\text{is } \sqrt{\frac{2}{\pi}} \left(\frac{1 - \cos as}{s^2} \right).$$

$$\text{Hence show that } \int_0^\infty \left(\frac{\sin t}{t} \right)^2 dt = \frac{\pi}{2}$$

3) Show that the Fourier Transform of

$$f(x) = \begin{cases} \frac{\sqrt{2\pi}}{2a} & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$$

$$\text{is } \frac{\sin sa}{sa}$$



4) Find the Fourier cosine Transform of e^{-ax} .

5) Find Fourier transform of

$$F(x) = \begin{cases} x^2, & |x| < a \\ 0, & |x| > a \end{cases}$$

6) Find Fourier Sine Transform of

$$f(x) = \frac{1}{x(x^2 + a^2)}$$

7) Find the Fourier Sine and Cosine Transform of $ae^{-\alpha x} + be^{-\beta x}$, $\alpha, \beta > 0$

8) Find $f(x)$ if its Fourier Sine transform is $\frac{s}{1+s^2}$

9) Find $f(x)$ if its Fourier Sine Transform is $(2\pi s)^{\frac{1}{2}}$



Questions Answer

$$1) \frac{1}{\sqrt{2\pi}} \frac{2i}{s^2}$$

$$4) F_c\{f(x)\} = \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + s^2} \right)$$

$$5) \left(\frac{2a^2}{s} - \frac{4}{s^2} \right) \sin as + \frac{4a}{s^2} \cos as$$

$$6) \frac{\pi}{2a^2} (1 - e^{-ax})$$

$$7) \frac{as}{s^2 + \alpha^2} + \frac{bs}{s^2 + \beta^2}, \quad \frac{a\alpha}{s^2 + \alpha^2} + \frac{b\beta}{s^2 + \beta^2}$$

$$8) \frac{2\sin^2 ax}{\pi^2 x^2}$$

$$9) \frac{1}{x\sqrt{x}}$$